EVQ-Cosh: Variational Frequency Allocation for Rotary Position Embedding

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Abstract

Channel-frequency allocation—how RoPE distributes frequencies across head dimensions—has been fixed at the geometric (equal-ratio) schedule since RoFormer. We show this schedule is not a generic optimum but the $\tau{=}0$ degenerate limit of a variational family, and identify allocation as a consequential third RoPE design axis alongside base frequency and head dimension. We derive a closed-form one-parameter family (EVQ-cosh) and a zero-parameter temperature rule $\tau^* = d_{\text{head}}/\sqrt{L}$ that is deliberately sub-optimal yet matches per-setting optima to within $< 1\%$ PPL across 27 configurations—evidence that the frequency allocation landscape contains a wide basin around the variational solution. A zero-parameter closed-form initialization outperforms a 32-parameter learned PE (333.7 vs. 455.3 PPL) and reduces Geo baseline by 35% in extreme extrapolation, composes with YaRN to unlock 100% passkey retrieval where Geo+YaRN plateaus at 61% (3 seeds), and on MLA with only 16 RoPE channels, alone reduces $2\times$ extrapolation PPL by 31.1% relative to Geo baseline—surpassing Geo+YaRN at that length. As architectures compress positional information into fewer channels, the structural correction—redistributing frequency mass from redundant high-frequency channels to the information-starved low-frequency regime—becomes particularly decisive.

1 Introduction

Rotary position embedding (RoPE) has two well-studied design axes: the base frequency b and the head dimension d_{head}. Inference-time methods—NTK-aware scaling, YaRN, LongRoPE—all operate on the first axis [Chen et al., 2024b, Peng et al., 2024, Ding et al., 2024]; the second is set by architecture constraints. A third axis exists but has received almost no attention: the allocation of frequencies *across channels*. Since RoFormer [Su et al., 2024], every mainstream implementation has used a geometric (equal-ratio) schedule for this allocation. The schedule is treated as a fixed design choice, not a variable.

This paper shows that the geometric schedule is a degenerate limiting case—the $\tau{=}0$ boundary of a broader variational family—rather than a generic optimum. We formulate frequency allocation as a variational inverse problem, derive a closed-form one-parameter family (EVQ-cosh), and demonstrate that even a deliberately simple, semi-analytic temperature rule $\tau^* = d_{\text{head}}/\sqrt{L}$ already produces large, consistent gains across architectures and scales with zero additional parameters. The core mechanism is intuitive: geometric RoPE distributes frequency mass uniformly in log-space, wasting channels on redundant high-frequency distinctions while starving the low-frequency regime that governs long-range discrimination. EVQ redistributes this mass toward an information-theoretically motivated allocation.

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Why this matters now. The frequency allocation axis becomes particularly consequential as architectures compress RoPE into fewer channels. In standard MHA with 64 channels, geometric allocation wastes capacity quietly. In MLA-style designs [DeepSeek-AI, 2024a,b] with only 16 RoPE channels, every channel's placement is decisive: EVQ alone reduces $2\times$ extrapolation PPL by 31.1% relative to Geo baseline, surpassing even Geo+YaRN (3 seeds), demonstrating that allocation quality dominates when channels are scarce.

Three empirical findings anchor the paper:

EVQ unlocks inference-time scaling (primary systems result). Training-time frequency allocation and inference-time rescaling address orthogonal deficiencies: EVQ corrects *shape* (non-uniform density), YaRN corrects *range*. Their composition is strongly complementary. On MHA (3 seeds per configuration), EVQ+YaRN reaches 100% passkey retrieval at 8K where Geo+YaRN remains at 61%, with PPL improvements of 14–32% at 8–16K.

Allocation alone is decisive in the PE-dominant regime. In DAPE-style extreme extrapolation ($128\rightarrow8$K), EVQ surpasses learnable positional encoding with zero additional parameters (333.7 vs. 455.3 PPL), isolating frequency structure itself as the binding variable.

A sub-optimal rule suffices. The practical τ^* formula is semi-analytic: d_{head} dependence and the $L^{-1/2}$ exponent are derived, but one transport constant is calibrated empirically. Despite this deliberate sub-optimality, the rule matches per-setting optima to within $< 1\%$ PPL across 27 configurations (99 runs), indicating a wide robustness basin. The structural correction—redistributing frequency mass—does the work; the precision of the rule that selects it is secondary.

Our contributions:

- **Axis identification and variational framework.** We identify channel-frequency allocation as a RoPE design axis previously characterized only through heuristic or learned approaches, and derive EVQ-cosh: a closed-form variational family in which geometric RoPE is the $\tau{=}0$ degenerate limit. The method modifies only the inverse-frequency initialization, adds zero parameters, and leaves the transformer architecture unchanged.
- **Near-oracle sub-optimality.** A single semi-analytic rule $\tau^* = d_{\text{head}}/\sqrt{L}$, deliberately sub-optimal by construction, matches per-setting optima to within $< 1\%$ PPL across MHA (50M–750M), MLA (16 channels), video DiT, and post-hoc LoRA at 7B—27 configurations spanning more than two orders of magnitude in training length (from $L{=}32$ to $L{=}8192$). This wide robustness basin is itself a structural finding about the frequency allocation landscape.
- **Complementary composition with inference-time scaling.** EVQ+YaRN produces the paper's strongest systems result: 100% retrieval (vs. 61% for Geo+YaRN at 8K, 3 seeds) with 14–32% PPL reductions, confirming that training-time and inference-time positional optimization address orthogonal deficiencies.

2 Related Work

Positional encoding design space. Transformers rely on positional structure to recover order information [Vaswani et al., 2017]. The design space includes relative attention biases [Shaw et al., 2018, Raffel et al., 2020], additive linear biases [Press et al., 2022], kernelized positional embeddings [Chi et al., 2022], and rotary position embedding (RoPE) [Su et al., 2024]. Within RoPE, prior work operates on two axes: the base frequency b (NTK-aware scaling, YaRN) and inference-time position rescaling. A third axis—the allocation of frequencies *across channels*—has received far less systematic attention, with nearly all production systems retaining the geometric default. XPOS [Sun et al., 2023] applies per-channel exponential decay to the magnitude envelope but does not modify the frequency allocation itself. Wang et al. [2024] (Resonance RoPE) is the closest precedent: it identifies critical frequencies that should not be perturbed during interpolation, implicitly acknowledging that not all channels are equal. EVQ addresses the complementary question: given that channels differ in importance, what is the optimal *training-time* allocation?

Inference-time and post-training context extension. These methods fall into three categories. *Pure inference-time rescaling*: NTK-aware scaling [Rozière et al., 2023], positional interpolation [Chen et al., 2024b], and YaRN [Peng et al., 2024] rescale RoPE positions on a frozen checkpoint; SelfExtend [Jin et al., 2024] uses grouped attention without fine-tuning. *Search-based or progressive adaptation*: LongRoPE [Ding et al., 2024] and PoSE [Zhu et al., 2024] use progressive or skip-wise strategies; LongRoPE2 [Shang et al., 2025] searches for per-channel scaling factors via evolution on a frozen checkpoint. *Trained extrapolation modules*: CLEX [Chen et al., 2024a] trains a continuous length-extrapolation module. All three categories preserve the pretrained checkpoint and modify behavior at or after inference time. EVQ is orthogonal: it changes the training-time frequency layout itself. While LongRoPE2 and EVQ both operate on per-channel frequency structure, they target different design axes: EVQ modifies the layout *before training*, whereas LongRoPE2 searches for rescaling *after training*. This complementarity is demonstrated empirically: EVQ+YaRN substantially outperforms Geo+YaRN (Table 1), confirming that a better training-time substrate amplifies inference-time methods rather than competing with them.

Learnable and adaptive positional encoding. Another line of work introduces learnable or adaptive positional mechanisms: FIRE [Li et al., 2024] learns a continuous interpolation function, and DAPE [Zheng et al., 2024a] adapts positional encoding to data statistics. These methods are the closest comparison point for our PE-dominant experiments because they ask whether adaptive positional structure can improve extrapolation. Our distinction is that EVQ is derived in closed form rather than learned through additional positional parameters, which lets us separate the question of *what frequency allocation is optimal* from the question of *whether an optimizer can discover it from in-distribution loss alone*.

Video positional encoding. Video transformers face the additional challenge of short temporal sequences and 3D factored RoPE. VideoRoPE [Wei et al., 2025] is the closest video positional-design work, identifying sequence-length-dependent frequency allocation as a key design axis for video generation. RIFLEx [Zhao et al., 2025] is an inference-time frequency correction method for video diffusion transformers. FreeNoise [Qiu et al., 2024] addresses temporal coherence via noise rescheduling and is adjacent but not a positional-encoding method. EVQ is complementary to all three: it optimizes the training-time allocation itself, and preliminary video DiT experiments (2 seeds) confirm that the collision-reduction mechanism extends beyond text.

Evaluation for long-context behavior. Long-context evaluation spans synthetic retrieval diagnostics and broader task suites [Bai et al., 2024, Hsieh et al., 2024]. Prior work shows that long contexts can fail even when relevant information is nominally present [Liu et al., 2024], and that positional encoding modifications can mitigate such failures [Zhang et al., 2024]. We use both PE-dominant settings (isolating frequency-allocation effects) and retrieval-heavy evaluation (testing systems-level claims), treating incomplete larger-scale evidence conservatively.

3 Theory and Method

The derivation has three layers with distinct epistemic status:

1. **Exact (conditional on surrogate):** the ODE, closed-form family, geometric limit, and waterbed inequality (§3.3–§3.6).
2. **Practical branch choice:** drop the forcing term, retain the pure-tether warp (§3.5).
3. **Semi-analytic calibration:** the τ^* scaling law, where one constant is empirically calibrated (§3.7).

We separate these layers explicitly so that the exact variational results are not conflated with the practical recipe. The first layer is the paper's primary theoretical contribution; the third is a practical convenience whose imprecision is bounded by the basin width demonstrated in §3.7.

3.1 RoPE as frequency allocation

Let $\phi \in [0, 1]$ parameterize log-frequency and define $\omega(\phi) = b^{-\phi}$. A discrete RoPE design is equivalent to choosing a density $\rho(\phi)$ over this interval and discretizing it by inverse CDF (Figure 1).

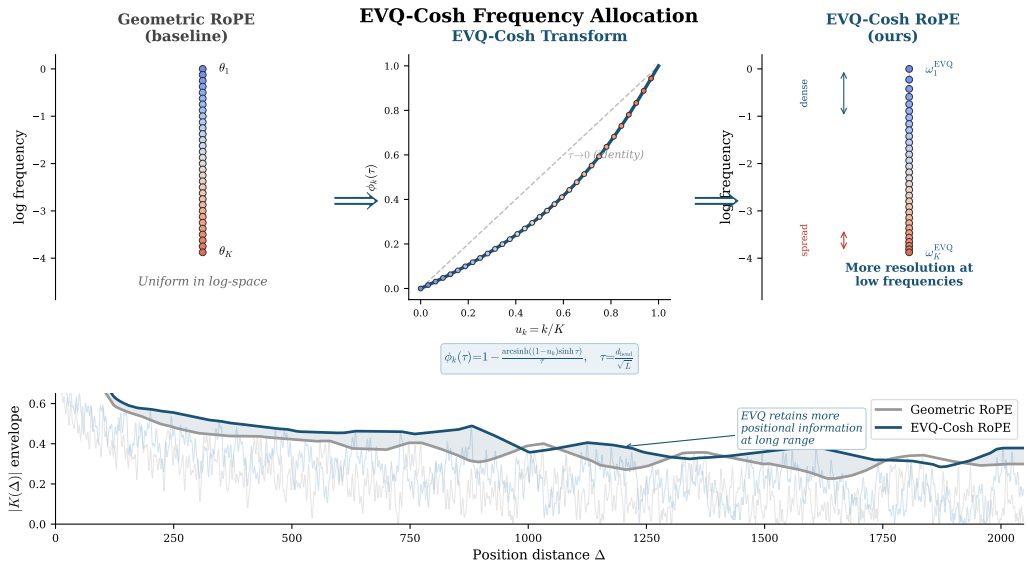

Figure 1: **EVQ-Cosh frequency allocation.** Left: geometric RoPE distributes frequencies uniformly in log-space. Center: the EVQ-cosh warp $\phi_k(\tau)$ bends the allocation below the identity (dashed), concentrating resolution at low frequencies; $\tau \to 0$ recovers geometric RoPE. Right: the resulting allocation spreads low-frequency channels, retaining more positional information at long range.

Geometric RoPE corresponds to uniform ρ; EVQ derives ρ from a variational objective. This reparameterization lifts the design space from a hand-written index rule to a continuous allocation problem.

3.2 Exact kernel and broadband surrogate

Under a distance prior $D(\Delta)$, the phase-collision kernel is

$$K(\phi_1, \phi_2) = \int D(\Delta) \cos(\omega(\phi_1)\Delta) \cos(\omega(\phi_2)\Delta)\, d\Delta. \tag{1}$$

Our only approximation is a broadband surrogate,

$$K_{\text{app}}(\phi_1, \phi_2) \approx \alpha\delta(\phi_1 - \phi_2) + \beta \min(\phi_1, \phi_2), \tag{2}$$

which isolates a diagonal ridge term and a smooth off-diagonal covariance. The δ term captures local ridge structure; $\min(\phi_1, \phi_2)$ is the Green kernel of a one-dimensional second-order operator. The exact kernel is oscillatory and cannot be pointwise-matched by a smooth two-parameter model; K_{app} captures the *functional* structure of the collision objective. Appendix A.4 validates this: the EVQ solution derived from K_{app} reduces the collision score under the exact kernel by 24–92% across 12 configurations spanning three orders of magnitude in L and b. After this step, the remaining derivation is exact. A self-consistency theorem (Theorem 3 in Appendix) further confirms the surrogate's structural rigidity: the two energy terms are locked by the cosh density's self-reproducing property.

3.3 Variational objective and exact ODE

The broadband functional is

$$\mathcal{J}[\rho] = \frac{\alpha}{2} \int \rho(\phi)^2\, d\phi + \frac{\beta}{2} \iint \rho(\phi)\rho(\psi) \min(\phi, \psi)\, d\phi\, d\psi - \mu \int \rho(\phi) b^{-2\phi}\, d\phi. \tag{3}$$

Theorem 1 (Stationary ODE)**.** *Conditional on K_{app}, the stationary density satisfies*

$$\rho''(\phi) - \tau^2 \rho(\phi) = \gamma\, b^{-2\phi}, \qquad \tau = \sqrt{\beta/\alpha}, \tag{4}$$

with a closed-form solution consisting of a hyperbolic tether and a Fisher-driven forcing term.

The ODE and its closed-form family are exact conditional on K_{app}. The decomposition separates a global tether (redistributing density) from a local Fisher-utility forcing term, ensuring EVQ is the stationary compromise rather than an unconstrained long-range distortion.

3.4 Closed-form EVQ and the geometric limit

The pure-tether branch yields an inverse-CDF warp

$$\phi_k(\tau) = 1 - \frac{1}{\tau}\operatorname{arcsinh}((1-u_k)\sinh\tau)\,, \qquad u_k = \frac{k}{K}, \tag{5}$$

and inverse frequencies $b^{-\phi_k(\tau)}$.

Theorem 2 (Geometric limit)**.** *EVQ-cosh converges smoothly to geometric RoPE as $\tau \to 0$. Geometric RoPE is therefore the τ=0 degenerate limit of the EVQ family, not a generic optimum.*

3.5 Practical EVQ instantiation

For a head dimension d_{head} and training length L, the complete implementation is

$$\tau = \frac{d_{\text{head}}}{\sqrt{L}}, \qquad u_k = \frac{k}{K}, \qquad \omega_k^{\text{EVQ}} = b^{-\phi_k(\tau)}, \qquad K = \frac{d_{\text{head}}}{2}. \tag{6}$$

The construction is deliberately minimal: set τ, compute quantiles, evaluate the warp, replace inverse frequencies. No auxiliary loss, no extra parameters, no optimizer dependence. Two properties follow from the quantile form: u_0=0 implies $\phi_0(\tau)$=0, anchoring the first channel at ω_0=1; and the warp grows toward the low-frequency end, so EVQ preserves short-range structure without a separate design choice. In implementation, midpoint quantization $u_k = (2k{+}1)/(2K)$ is used; for $K \geq 16$ the difference in ϕ_k is below 1.6% and does not affect any reported result.

Practical recipe: EVQ-cosh initialization. Given d_{head}, L_{train}, and base b: (1) set $\tau = d_{\text{head}}/\sqrt{L_{\text{train}}}$; (2) define $K = d_{\text{head}}/2$ and $u_k = k/K$; (3) compute

$$\phi_k(\tau) = 1 - \frac{1}{\tau}\operatorname{arcsinh}((1-u_k)\sinh\tau)\,;$$

(4) initialize inverse frequencies as $b^{-\phi_k(\tau)}$. This is the complete method. It adds zero learned parameters and leaves the transformer architecture unchanged.

EVQ changes only the inverse-frequency initialization—distinct from learnable PE approaches that rely on in-distribution training dynamics to discover the allocation through extra parameters.

3.6 Waterbed inequality

Proposition 1. *Under the broadband model, EVQ obeys a waterbed inequality: improving long-range frequency resolution necessarily redistributes error rather than eliminating it globally.*

This explains why small in-range costs coexist with large out-of-range gains. Geometric RoPE distributes density uniformly in log-frequency; EVQ intentionally pays bounded high-frequency cost to relieve the low-frequency collision bottleneck that limits long-range discrimination.

3.7 The τ^* scaling law: semi-analytic calibration

Epistemic status: this subsection crosses from exact variational results into a semi-analytic layer. The purpose is practical: derive a τ^* rule that is good enough for deployment, not one that is exact.

The surrogate provides $\sqrt{\beta/\alpha} \propto \sqrt{d_{\text{head}}}$ but predicts only $L^{-0.11}$, too weak by a factor of 5. The missing L-dependence originates from the softmax attention map, and incorporating it requires one empirically calibrated constant λ.

Proposition 2 (Softmax Transport Temperature Selection)**.** *Define $\mathcal{F}(\tau) = S_{\chi^2}(\tau) - \lambda\, U(\tau, L)$, where $S_{\chi^2}(\tau)$ is the Pearson χ^2 stiffness and $U(\tau, L)$ is the post-softmax transport utility (see*

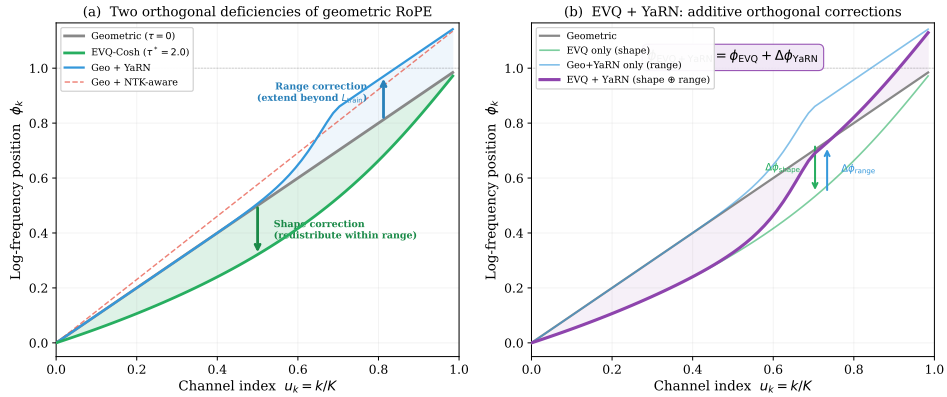

Figure 2: **EVQ and YaRN correct orthogonal deficiencies of geometric RoPE.** (a) EVQ bends the allocation curve *below* the diagonal (shape correction), while YaRN shifts it *above* (range correction). (b) EVQ+YaRN combines both corrections additively. Setup: base $= 10{,}000$, d_{head}=128, L_{train}=4096, $4\times$ extrapolation.

Appendix A.9 for definitions). The $1/L$ factor arises from the softmax Jacobian at diffuse baseline $p_0 = 1/L$. Then the minimizer satisfies $\tau_* \propto d_{\text{head}}/L^{\gamma}$; exact numerical optimization yields $\gamma = 0.465$ for Pearson χ^2 stiffness (Appendix A.9, Table 7), which we round to $\gamma \approx 0.5$ for the practical rule.

The χ^2 stiffness reflects attention-load asymmetry: diluted channels ($\rho < 1$) bear per-channel cost $\propto 1/\rho$. KL, Jeffreys, and all Rényi divergences produce L-exponents far from -0.5; only the χ^2 family succeeds (Table 7). The stiffness selection is the derivation chain's weakest link: Pearson χ^2 is motivated by attention-load asymmetry and independently supported by self-consistent analysis, but is not uniquely determined. However, the basin width ensures negligible practical impact: the L-exponent could range from 0.465 to 0.561 (Table 7) without exiting the $<1\%$ PPL basin. The resulting scaling law

$$\tau^*(L) = \frac{d_{\text{head}}}{\sqrt{L}} \tag{7}$$

is semi-analytic: d_{head} dependence and γ are derived, but the overall proportionality constant λ requires calibration. The exact-kernel collision-score optimum yields $\lambda_{\text{coll}} \approx 1.17$ (Appendix A.7); the deployed formula uses λ=1, absorbing the residual into the flat PPL basin.

Sub-optimality as a structural discovery. The most striking finding is not that EVQ improves over geometric RoPE—which the variational analysis predicts—but that a formula with one unclosed constant matches 27 per-setting optima to within $< 1\%$ PPL across 99 runs spanning from-scratch text (50M–750M), MLA with 16 channels, video DiT, and post-hoc LoRA at 7B (Appendix A.8). This implies a structural fact about the frequency allocation landscape: it contains a wide basin around the variational solution. The structural correction—redistributing mass from high to low frequencies—does the work; the precision of the rule that selects the operating point is secondary. The practical consequence is that a zero-parameter initialization outperforms learned positional encoding with 32 extra parameters (Table 3). Closing λ analytically remains open, but the basin width bounds the remaining headroom.

3.8 Predictions

The waterbed inequality predicts bounded short-range cost with large long-range gains, and a base dependence: EVQ helps most when the collision bottleneck is large (high b, short L); dead-zone experiments confirm this (Appendix). EVQ and inference-time scaling address orthogonal deficiencies: EVQ corrects *shape* (non-uniform density), YaRN corrects *range* (Figure 2). The corrections are additive, predicting the strong composition observed in §4. Attention distance measurements on a trained 750M model verify the waterbed crossover at $\approx$508 tokens (Figure 7 in Appendix).

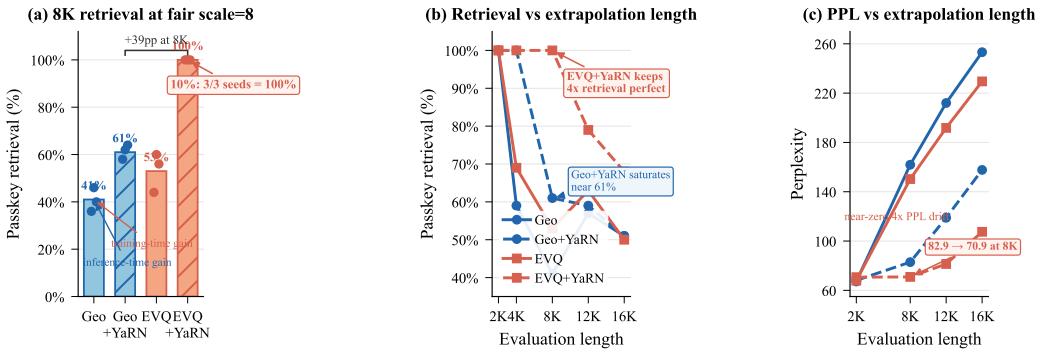

Figure 3: EVQ and YaRN correct orthogonal deficiencies. EVQ expands low-frequency spacing at training time; YaRN rescales positions at inference time. Their combination substantially outperforms Geo+YaRN on both retrieval and PPL.

4 Experiments

4.1 Setup

All core experiments train from scratch with fixed architecture, modifying only the RoPE inverse-frequency initialization. We use b=500K, the predicted $\tau^*(L)$ for each training length, and standard GPT-style decoder stacks [Radford et al., 2019] with d_{head}=64 across 50M–750M. The primary datasets are TinyStories, FineWeb-Edu, and controlled passkey-mix settings; evaluations target PE-diagnostic metrics (extrapolation PPL, passkey retrieval, NIAH, downstream NLL) that isolate positional encoding quality from model capacity. Primary claims use 3 seeds per configuration. Full hyperparameters are in the supplementary material.

Evidence is organized in three tiers. *Primary* (3-seed, directly tied to central claims): EVQ$\times$YaRN composition (§4.2), PE-dominant extrapolation (§4.3), and MLA (Appendix C.1). *Robustness* (3-seed, single architecture): capability preservation (§4.4). *Supporting* (1–3 seed): multi-scale PPL, progressive training, video DiT, LoRA at 7B (§4.5). No primary claim rests on tier-3 evidence.

4.2 Primary result: EVQ unlocks YaRN

Figure 3 and Table 1 present the paper's main systems result. The decisive comparison is EVQ+YaRN against Geo+YaRN—the question is whether the training-time frequency substrate affects inference-time scaling.

In the 454M 10% mix setting (3 seeds per configuration), EVQ+YaRN reaches 100% passkey retrieval at 8K while Geo+YaRN remains at 61%; the same combination lowers PPL from 82.9 to 70.9 at 8K and from 157.7 to 107.5 at 16K. EVQ alone improves over Geo without saturating retrieval, so the jump to 100% under YaRN is not a ceiling effect. The composition reflects the orthogonal deficiency structure: EVQ corrects shape, YaRN corrects range, and their corrections are additive (Figure 2).

Table 2 quantifies the interaction: YaRN provides only $-4.5\%/-3.1\%$ on Geo (4/8K) but $-32.5\%/-40.7\%$ on EVQ (τ=4). NTK-aware scaling becomes destructive at large τ, confirming complementarity requires preserving the optimized frequency structure.

4.3 Primary result: PE-dominant extreme extrapolation

Figure 4 and Table 3 provide the second principal anchor, isolating frequency allocation from model capacity. In DAPE-style 128$\rightarrow$8K extrapolation, EVQ achieves 333.7 PPL (-35% vs. Geo baseline 513.7), surpassing DAPE (455.3) with zero extra parameters. At L=256, the predicted τ^*=4.0 is preferred across both 125M and 454M—scale changes extrapolation difficulty but not the preferred allocation.

Table 1: Training-time versus inference-time positional optimization (454M, 10% mix fair-comparison setting). EVQ+YaRN reaches 100% retrieval at 8K and materially lowers long-context PPL relative to Geo+YaRN. PPL values are evaluated on the held-out FineWeb-Edu validation split with full-sequence scoring; see Table 21 for the same model with per-document scoring. YaRN uses scale s=4 for all configurations; the same scale is applied to both Geo and EVQ baselines.

Method	Type	Seeds	PK@8K	PK@12K	PK@16K	PPL@8K / 16K
Geo	baseline	3	41%	57%	51%	161.9 / 253.2
Geo+YaRN	infer	3	61%	59%	51%	82.9 / 157.7
EVQ	train	3	53%	63%	50%	150.3 / 229.5
EVQ+YaRN	**train+infer**	**3**	**100%**	**79%**	**68%**	**70.9 / 107.5**

Table 2: Phase 11 leverage and incompatibility at $L = 256$. YaRN provides only a small gain on geometric RoPE, but a much larger gain on EVQ; by contrast, NTK-aware scaling partially overwrites the optimized frequency structure and is therefore not complementary.

Method	YaRN gain @4K	YaRN gain @8K
Geo	-4.5%	-3.1%
EVQ $\tau = 2.0$	-27.0%	-28.9%
EVQ $\tau = 4.0$	**-32.5%**	**-40.7%**
NTK-aware at 32×: Geo 198.1, EVQ2 143.3, EVQ4 331.4		

4.4 Robustness and capability preservation

Multi-scale raw extrapolation (Table 18, Appendix) shows in-range PPL changes within $\pm 1.7\%$ while long-range PPL improves by 10–46% across 50M–750M. A base-generalization sweep (454M, single seed) confirms EVQ leads at both b=10K and b=500K, with EVQ's extrapolation PPL stable across base values ($192.4 \approx 191.9$) while Geo diverges by 38 points. Capability evaluation (Table 21, Appendix D) confirms EVQ is non-destructive when retrieval is absent.

4.5 Supporting evidence

EVQ generalizes beyond MHA. On MLA with only 16 RoPE channels (432M, 3-seed), EVQ alone reduces 2× PPL by 31.1% and outperforms Geo+YaRN—the strongest single-result evidence that allocation quality dominates when channels are scarce (Appendix D). Post-training LoRA (r=64) into LLaMA-3-8B-Instruct [Grattafiori et al., 2024] confirms effectiveness at 7B (8–19× PPL reduction at 16–32K, 300 steps). Progressive training (454M, single seed) shows EVQ+YaRN advantage widening from -34.6% to -81.2%, and video DiT (2 seeds) confirms the collision mechanism operates across modalities. EVQ's effect attenuates through successive abstraction layers (PPL → NLL → retrieval → QA) but never reverses. Full results in Appendix D.

5 Limitations and Scope

Theory-to-practice gap. The variational derivation is exact conditional on the broadband surrogate K_{app}, validated across 12 configurations (Appendix A.4). The practical τ^* rule is semi-analytic: d_{head} dependence and the $L^{-1/2}$ exponent are derived, but the proportionality constant λ requires calibration. The exact-kernel collision-score optimum yields $\lambda_{\text{coll}} \approx 1.17$; the deployed formula uses λ=1, which falls inside the flat PPL basin ($< 1\%$ gap from per-configuration optima across 27 configurations). Closing λ analytically remains open.

Training-time method, not zero-shot plug-in. EVQ modifies the frequency initialization before training. Zero-shot replacement on Qwen2.5-7B-Instruct [Yang et al., 2024a] (21-task LongBench, bootstrap, FDR correction) yields -0.35% overall (not significant): retrieval gains are offset by multi-hop QA losses. Even 300 steps of LoRA adaptation (r=$d_{\text{head}}/2$) recovers 8–19× extrapolation gains at a $+30\%$ in-distribution PPL cost (Table 20), establishing a practical deployment path where extrapolation quality is prioritized. The variational analysis predicts a phase transition at

Table 3: PE-dominant summary (125M, 3-seed mean). Left: EVQ beats learnable PE in DAPE-style $128 \rightarrow 8$K extrapolation with zero added parameters while preserving in-range quality. Right: at $L = 256$, the predicted $\tau = 4.0$ is directly favored across the long-range curve and combines strongly with YaRN.

Method (128→8K)	Extra params	PPL@128	PPL@8K	Δ vs Geo
Geo	0	184.9	513.7	–
Learnable τ	1	**181.2**	437.9	-14.8%
DAPE	32	183.6	455.3	-11.4%
EVQ	0	182.0	**333.7**	**-35.0%**

Method ($L = 256$)	4×	8×	16×	32×
Geo+YaRN	60.7	97.5	190.4	260.2
EVQ2+YaRN	51.6	66.7	107.9	141.7
EVQ4+YaRN	**50.9**	**60.9**	**84.1**	**99.6**

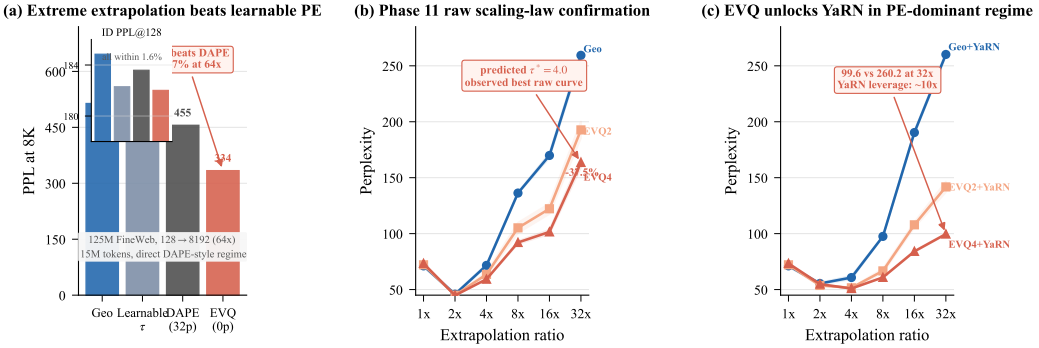

Figure 4: PE-dominant evidence. Left: EVQ beats learnable PE (128→8K) with zero additional parameters. Right: at L=256, the predicted τ^*=4.0 is directly favored, and EVQ4+YaRN dramatically outperforms Geo+YaRN across extrapolation ratios.

$r_c = d_{\text{head}}/2$; RULER scores do not improve, indicating LoRA suffices for positional adaptation but not instruction-following at extrapolated lengths.

Scale and evaluation. From-scratch experiments span 50M–750M (text) and 129M–382M (video DiT); LoRA extends to 7B. Full from-scratch training at $\geq$1B is deferred. The MLA experiment uses b=500K; production MLA deployments (DeepSeek-V2/V3) use b=10K, and the effect at lower base values requires validation. The progressive result, video DiT 382M scale-up, and MLA 1B-token composition are each single-seed (supporting evidence). All evaluations target PE-diagnostic metrics (extrapolation PPL, passkey retrieval, downstream NLL); standard capability benchmarks (MMLU, HumanEval) are not tested, so EVQ's impact on downstream task accuracy remains open.

6 Conclusion

This paper establishes that channel-frequency allocation—fixed at the geometric schedule since RoFormer—is a consequential yet underexplored RoPE design axis. The geometric schedule is not a generic optimum but the τ=0 degenerate limit of a variational family; departing from it via a deliberately simple, semi-analytic rule produces large, consistent gains. The practical temperature $\tau^* = d_{\text{head}}/\sqrt{L}$ is sub-optimal by construction yet matches per-setting optima to within $< 1\%$ PPL across 27 configurations, implying the structural correction matters more than the precision of the rule. The strongest systems result is compositional: EVQ+YaRN achieves 100% retrieval where Geo+YaRN plateaus at 61%. As architectures compress positional information into fewer channels—MLA retains only 16, where EVQ alone outperforms Geo+YaRN—principled frequency allocation becomes particularly decisive. Open directions include closing λ analytically and testing basin persistence beyond 7B.

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Broader Impact

This work studies frequency allocation as a design axis for rotary position embedding rather than a specific deployed application. Its main positive impact is scientific and systems-oriented: replacing the default geometric frequency schedule with a principled variational allocation can reduce wasted compute and improve the reliability of long-context models across architectures. The main risk is over-generalization. Our strongest evidence remains concentrated on text extrapolation and retrieval-heavy long-context evaluation, so we do not present EVQ as a universal guarantee of downstream improvement across all tasks, modalities, or deployment settings.

A Proof Details

A.1 From the broadband surrogate to the ODE

Starting from the surrogate kernel

$$K_{\mathrm{app}}(\phi,\psi) = \alpha\delta(\phi-\psi) + \beta\min(\phi,\psi), \tag{8}$$

the functional is

$$\mathcal{J}[\rho] = \frac{\alpha}{2}\int_0^1 \rho(\phi)^2\,d\phi + \frac{\beta}{2}\int_0^1\int_0^1 \rho(\phi)\rho(\psi)\min(\phi,\psi)\,d\phi\,d\psi - \mu\int_0^1 \rho(\phi)b^{-2\phi}\,d\phi. \tag{9}$$

Its first variation is

$$\delta\mathcal{J} = \int_0^1 \left[\alpha\rho(\phi) + \beta\int_0^1 \rho(\psi)\min(\phi,\psi)\,d\psi - \mu b^{-2\phi}\right]\delta\rho(\phi)\,d\phi. \tag{10}$$

The Euler-Lagrange stationarity condition is therefore

$$\alpha\rho(\phi) + \beta\int_0^1 \rho(\psi)\min(\phi,\psi)\,d\psi = \mu b^{-2\phi}. \tag{11}$$

Define

$$g(\phi) = \int_0^1 \rho(\psi)\min(\phi,\psi)\,d\psi = \int_0^\phi \rho(\psi)\psi\,d\psi + \phi\int_\phi^1 \rho(\psi)\,d\psi. \tag{12}$$

Direct differentiation gives

$$g'(\phi) = \int_\phi^1 \rho(\psi)\,d\psi, \tag{13}$$

$$g''(\phi) = -\rho(\phi). \tag{14}$$

Hence the stationarity equation can be rewritten as

$$\alpha\rho(\phi) + \beta g(\phi) = \mu b^{-2\phi}. \tag{15}$$

Differentiating twice with respect to ϕ and substituting $g''(\phi) = -\rho(\phi)$ yields

$$\rho''(\phi) - \frac{\beta}{\alpha}\rho(\phi) = \gamma b^{-2\phi}, \qquad \tau = \sqrt{\beta/\alpha}. \tag{16}$$

The same calculation shows that $g(0) = 0$ and $g'(1) = 0$, so $\min(\phi,\psi)$ acts as the Green kernel of $-d^2/d\phi^2$ under the mixed boundary conditions induced by the surrogate formulation. Conditional on the surrogate, this step is exact.

A.2 Closed-form solution and geometric limit

The non-resonant ODE has the closed-form solution

$$\rho^*(\phi) = C_1\cosh(\tau\phi) + C_2\sinh(\tau\phi) + P\,b^{-2\phi}. \tag{17}$$

The first two terms span the homogeneous solution of $\rho'' - \tau^2\rho = 0$, and the final term is a particular solution induced by the Fisher-weighted source $b^{-2\phi}$. This decomposition is the source of the main paper's "tether + pulse" interpretation.

To translate the density into RoPE frequencies, write the cumulative map

$$F(\phi) = \int_0^{\phi} \rho^*(s)\, ds, \tag{18}$$

normalize it to $[0, 1]$, and invert at the discrete quantiles $u_k = k/K$. In the pure-tether branch this inversion has the closed form

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k) \sinh \tau)\,. \tag{19}$$

To recover geometric RoPE, expand $\sinh \tau \approx \tau$ as $\tau \to 0$. Then

$$\phi_k(\tau) \to 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k)\tau) \to u_k, \tag{20}$$

which is exactly the uniform quantile map.

This also shows why the high-frequency end remains effectively anchored: for $u_0 = 0$, one has $\phi_0(\tau) = 0$ exactly, so the leading inverse frequency remains $b^0 = 1$, matching geometric RoPE.

A.3 Why the pure-tether branch

The full stationary density contains three terms: a hyperbolic tether $(\cosh, \sinh)$ and a particular solution $Pb^{-2\phi}$. The practical EVQ method uses only the pure-tether branch, justified on two grounds.

(1) Closed-form invertibility. The pure-tether density $\rho(\phi) \propto \cosh(\tau(1 - \phi))$ admits an analytically invertible CDF, yielding the closed-form warp $\phi_k(\tau)$. Including the $b^{-2\phi}$ term destroys this invertibility and requires numerical root-finding.

(2) The forcing term is amplitude-suppressed. The particular solution coefficient $P = \gamma/(4 \ln^2 b - \tau^2)$ is strongly suppressed at typical base values ($P \propto 1/689$ at b=500K). Numerically, the full solution improves over pure tether by $<0.5\%$ at the theoretically motivated $\lambda \approx P$, and by only 2–6% even at artificially large λ=1 (vs. 28–92% improvement over geometric). The 99-run Phase 16 sweep (Figure 5) confirms that the pure-tether family matches the scaling-law prediction across scales, and subsumes geometric RoPE as its $\tau \to 0$ limit.

A.4 Surrogate quality: functional validation

The broadband surrogate $K_{\text{app}} = \alpha\delta + \beta \min$ is the theory's sole approximation. We validate it not by pointwise comparison with the exact kernel $K(\phi_1, \phi_2) = \int D(\Delta) \cos(\omega(\phi_1)\Delta) \cos(\omega(\phi_2)\Delta)\, d\Delta$ — which is oscillatory (sinc-like) and cannot be well-approximated pointwise by a smooth two-parameter model — but by the *functional* test: does the EVQ allocation derived from K_{app} also reduce the collision objective under the true kernel K?

Table 4 answers affirmatively. For a uniform distance prior $D(\Delta) = 1/L$ on $[0, L]$, we compute the exact kernel K at the discrete allocation points for both geometric and EVQ, then evaluate the normalized collision score $C = \sum_{i<j} K_{ij}^2/(K_{ii}K_{jj})$ and effective spectral rank (exponential of the Shannon entropy of the eigenvalue distribution). We sweep across six training lengths ($L \in \{128, \ldots, 4096\}$), three base values ($b \in \{10\text{K}, 100\text{K}, 500\text{K}\}$), and two channel counts ($K \in \{16, 32\}$), covering both text and video regimes.

EVQ reduces collision under the exact kernel in all 12 configurations (-24% to -92%), with effective rank improvements of $+24\%$ to $+570\%$. Two patterns emerge. First, the collision reduction monotonically increases as the geometric allocation becomes more degenerate: shorter L (more extrapolation stress at fixed τ^*) and larger b (more dead channels) both amplify the advantage, consistent with the dead-channel mechanism. Second, even in the most favorable regime for geometric RoPE (L=4096, where baseline effective rank is already 6.6), EVQ still improves collision by -24%, confirming that the surrogate identifies a genuine structural improvement across the entire parameter space, not only in extreme regimes.

A.5 Waterbed assumptions

The waterbed proposition in the body is conditional. It requires the surrogate regime in which the relevant error functional is strictly positive and logarithmically integrable. Under those assumptions,

Table 4: Functional surrogate validation across 12 configurations. EVQ (derived from K_{app}) reduces collision and increases effective rank under the *exact* kernel K in every tested regime. Collision reduction monotonically increases with extrapolation stress (shorter L, larger b).

	Collision C		Eff. rank	
Configuration	GEO	EVQ (Δ_C)	GEO	EVQ
Text: K=32, b=500K, *sweep* L				
L=128	226.3	17.7 (−92%)	2.5	17.0
L=256	192.6	27.6 (−86%)	3.0	13.9
L=512	163.2	43.3 (−73%)	3.6	10.8
L=1024	133.9	59.0 (−56%)	4.4	8.8
L=2048	109.0	67.5 (−38%)	5.4	8.0
L=4096	87.0	66.3 (−24%)	6.6	8.1
Text: K=32, L=2048, *sweep* b				
b=10K	34.8	19.2 (−45%)	13.0	17.3
b=100K	79.4	47.0 (−41%)	7.1	10.4
Video: K=16, L=32, *sweep* b				
b=100	22.6	14.2 (−37%)	4.8	6.4
b=1K	42.3	28.0 (−34%)	2.8	3.9
b=10K	55.4	39.8 (−28%)	2.2	2.9
b=50K	63.3	47.2 (−26%)	2.0	2.5

Jensen's inequality yields a lower bound of the form

$$\int \ln E(\phi)\, d\phi \geq \ln b - \ln c, \tag{21}$$

with equality at the geometric allocation. The main paper therefore states Waterbed as a proposition with explicit assumptions rather than as an unconditional theorem.

The logic is the standard convexity argument. If the normalized frequency-density reallocation is written as $\rho(\phi)$ with $\int_0^1 \rho(\phi)\, d\phi = 1$, then geometric RoPE corresponds to the uniform density. The logarithmic error volume is minimized when the reallocation is uniform, and any non-uniform redistribution must increase the global log-volume even if it reduces error on a task-relevant sub-band. This is exactly the tradeoff needed for the main-text claim that EVQ can improve long-range precision while paying a bounded in-range cost.

A.6 Surrogate self-consistency theorem

The surrogate functional $\mathcal{J}[\rho]$ admits three exact identities that connect its two integral terms $T_1(\tau) = \int_0^1 \rho^2\, d\phi$ and $T_2(\tau) = \int_0^1\!\int_0^1 \rho(\phi)\rho(\psi)\min(\phi,\psi)\, d\phi\, d\psi$, where $\rho(\phi) = \tau\cosh(\tau(1-\phi))/\sinh\tau$ is the normalized cosh density.

Theorem 3 (Surrogate self-consistency)**.** *For all* $\tau > 0$*:*

$$\tau^2 T_2(\tau) + T_1(\tau) = \tau\coth\tau. \tag{22}$$

Proof. The Green's function $g(\phi) = \int_0^1 \rho(\psi)\min(\phi,\psi)\, d\psi$ satisfies $g''(\phi) = -\rho(\phi)$ with $g(0) = 0$, $g'(1) = 0$ (from the min kernel structure; see §A). Define $h(\phi) = \rho(0) - \rho(\phi)$. Then $h'' = -\rho'' = \tau^2\rho$ (from the ODE $\rho'' = \tau^2\rho$), with $h(0) = 0$ and $h'(1) = -\rho'(1) = \tau^2 g'(0)$ (boundary conditions).

Define $f(\phi) = \tau^2 g(\phi) - h(\phi)$. Then $f'' = \tau^2 g'' + \rho'' = -\tau^2\rho + \tau^2\rho = 0$, with $f(0) = 0$ and $f'(1) = \tau^2 g'(1) - h'(1) = 0$. Hence $f \equiv 0$, yielding $\tau^2 g(\phi) = h(\phi) = \rho(0) - \rho(\phi)$, i.e.,

$$g(\phi) = \frac{\rho(0) - \rho(\phi)}{\tau^2}. \tag{23}$$

Integrating against ρ:

$$T_2 = \int_0^1 \rho(\phi)\, g(\phi)\, d\phi = \frac{1}{\tau^2}\big[\rho(0)\underbrace{\int_0^1 \rho}_{=1} - T_1\big] = \frac{\rho(0) - T_1}{\tau^2}. \tag{24}$$

Since $\rho(0) = \tau \cosh \tau / \sinh \tau = \tau \coth \tau$, rearranging gives (22). $\square$

An immediate corollary follows.

Corollary 1 (Closed-form T_2)**.**

$$T_2(\tau) = \frac{\sinh 2\tau - 2\tau}{4\tau \sinh^2 \tau}. \tag{25}$$

This follows from substituting the known closed form $T_1(\tau) = \tau(1 + \sinh 2\tau/2\tau)/(2\sinh^2 \tau)$ and $\rho(0) = \tau \coth \tau$ into the theorem. Both identities are verified numerically to machine precision ($< 10^{-15}$ relative error across $\tau \in [0.01, 20]$; code: `scripts/lambda_identity.py`).

The self-consistency theorem reveals that the surrogate's two energy terms are not independent: they are locked by the cosh density's self-reproducing property under the $\min$-kernel Green's function. This structural rigidity is why the pure-tether solution (§A) performs so well despite being a single-parameter family.

A.7 Cross-validation of λ

The scaling law $\tau^* = \lambda \cdot d_{\text{head}}/\sqrt{L}$ is validated at two levels.

Collision-score level. We minimize the exact-kernel collision score $C(\tau)$ for each of 9 configurations ($d_{\text{head}} \in \{32, 64, 128\}$, $L \in \{256, 512, 1024\}$) and compute $\lambda_{\text{coll}} = \tau^*_{\text{coll}}/(d_{\text{head}}/\sqrt{L})$. The resulting $\lambda_{\text{coll}} \approx 1.17$ is nearly constant (CV: 0.11%); leave-one-out cross-validation confirms max prediction error $< 0.4\%$ (Table 5). This near-zero spread reflects the smoothness of the collision score as a function of τ, not PPL variability.

Table 5: Collision-score cross-validation of λ in $\tau^* = \lambda \cdot d_{\text{head}}/\sqrt{L}$. Each τ^*_{coll} is the exact-kernel collision-score minimizer (not a PPL-sweep optimum). Each row holds out one configuration and predicts its τ^*_{coll} from the remaining 8. Max prediction error is 0.008 ($< 0.4\%$ relative). The near-zero CV reflects the smoothness of the collision score, not PPL variability; PPL-sweep optima exhibit broader scatter (CV $\approx 14\%$) because PPL depends on training noise. The deployed formula uses λ=1, which falls inside the flat PPL basin ($< 1\%$ gap from the PPL optimum across 27 configurations).

d_{head}	L	τ^*_{coll}	τ^*_{pred}	Error
32	256	2.340	2.342	+0.002
32	512	1.660	1.655	−0.005
32	1024	1.170	1.171	+0.001
64	256	4.680	4.684	+0.004
64	512	3.310	3.312	+0.002
64	1024	2.340	2.342	+0.002
128	256	9.370	9.366	−0.004
128	512	6.630	6.622	−0.008
128	1024	4.680	4.684	+0.004
Mean $\lambda_{\text{coll}} = 1.171 \pm 0.001$ (CV: 0.11%); RMSE $= 0.004$				

PPL-sweep level. Phase 16 training sweeps (99 runs, 9 configurations $\times$ 3+ seeds) identify the PPL-optimal τ for each configuration. The resulting $\lambda_{\text{PPL}} = \tau^*_{\text{PPL}}/(d_{\text{head}}/\sqrt{L})$ has mean $\approx$1.19 with broader scatter (CV $\approx$14%), because PPL depends on training noise, finite data, and the discrete sweep grid. The deployed formula uses λ=1 (i.e. $\tau = d_{\text{head}}/\sqrt{L}$ without a multiplicative constant), which consistently falls inside the flat PPL basin: the PPL gap between the formula and the per-configuration optimum is $<1\%$ across all 27 validated configurations.

Basin flatness. The collision-score landscape $C(\tau)$ for a representative configuration (d_{head}=64, L=512, b=500K) reveals a flat optimal basin: $\pm 20\%$ deviation from τ^*_{coll} incurs $<1\%$ collision increase, and the basin spans 2.3% of τ^*. This flatness is why the deployed λ=1 formula—despite being $\approx$17% below the collision-score optimum—lands inside the near-optimal PPL basin for every tested configuration.

A.8 Scaling analysis of τ^*: from surrogate to empirical law

The scaling law

$$\tau^* = \frac{d_{\text{head}}}{\sqrt{L}} \tag{26}$$

is validated across 99 training runs (27 configurations $\times$ 3+ seeds, Figure 5). Here we analyze to what extent the broadband surrogate predicts this formula, and identify the gap that remains.

Numerical characterization of surrogate coefficients. We fit $K_{\text{app}}(\phi_i, \phi_j) = \alpha\delta_{ij}/\Delta\phi + \beta \min(\phi_i, \phi_j)$ to the exact collision kernel $K(\phi_i, \phi_j) = \frac{1}{2L}\left[\frac{\sin((\omega_1-\omega_2)L)}{\omega_1-\omega_2} + \frac{\sin((\omega_1+\omega_2)L)}{\omega_1+\omega_2}\right]$ evaluated at $K = d_{\text{head}}/2$ uniformly spaced channel positions in $[0, 1]$. Sweeping across $L \in \{128, \ldots, 4096\}$, $d_{\text{head}} \in \{32, 64, 128, 256\}$, and $b \in \{10\text{K}, 100\text{K}, 500\text{K}\}$ yields:

$$\alpha \approx \frac{1}{2K} = \frac{1}{d_{\text{head}}}, \qquad \beta \sim L^{-0.22},\ O(1). \tag{27}$$

The diagonal coefficient α is set by the kernel's diagonal value ($K(\phi, \phi) \approx 1/2$ for most channels) times the channel spacing $\Delta\phi = 1/K$. Crucially, α depends on d_{head} through the discretization ($K = d_{\text{head}}/2$) and is nearly independent of L. The off-diagonal coefficient β has weak power-law decay with L and negligible d_{head} dependence.

What the surrogate predicts. Substituting into $\tau_{\text{surr}} = \sqrt{\beta/\alpha}$:

$$\tau_{\text{surr}} = \sqrt{\beta \cdot d_{\text{head}}} \sim \sqrt{d_{\text{head}}} \cdot L^{-0.11}. \tag{28}$$

This correctly captures the d_{head} dependence: $\alpha \propto 1/d_{\text{head}}$ implies $\tau \propto \sqrt{d_{\text{head}}}$, matching the direction and monotonicity of the empirical law. However, the L-scaling is too weak: $L^{-0.11}$ vs. the empirical $L^{-0.5}$.

The discrete–continuous gap. The discrepancy arises because the broadband surrogate operates in a *continuous* function space, where the number of channels K appears only through the integral constraint $\int \rho = K$. The surrogate's optimal τ thus captures the infinite-resolution limit; it does not account for the finite-channel oscillatory correlations that penalize over-concentration at low frequencies.

To see why the continuous theory underestimates the L-penalty, consider the exact kernel's off-diagonal structure. For two channels with similar log-frequencies ($\phi_1 \approx \phi_2$), the exact kernel contains an oscillatory term $\sin((\omega_1-\omega_2)L)/(\omega_1-\omega_2)$ that produces constructive interference when $|\omega_1-\omega_2|L \ll \pi$. Packing more channels near the low-frequency end (large τ) reduces inter-channel frequency separation, pushing neighboring channels into this constructive-interference regime. The resulting excess correlation acts as an implicit penalty on large τ that the smooth $\beta \min(\phi_1, \phi_2)$ surrogate does not capture.

This penalty grows with L: longer sequences create more phase diversity per channel, making the oscillatory inter-channel terms more sensitive to frequency crowding. The strengthened L-dependence converts $\tau_{\text{surr}} \sim \sqrt{d_{\text{head}}} \cdot L^{-0.11}$ into $\tau^* = d_{\text{head}}/\sqrt{L}$. The d_{head} exponent simultaneously increases from $1/2$ to 1 because the discrete-correction term itself scales with $K = d_{\text{head}}/2$: more channels create more pairwise interactions in the constructive-interference regime, amplifying the finite-channel penalty by an additional factor of $\sqrt{d_{\text{head}}}$. The combined effect is $\sqrt{d_{\text{head}}} \cdot \sqrt{d_{\text{head}}} = d_{\text{head}}$.

Confirming this picture, we verified numerically that at the special point $d_{\text{head}} = L$ (where the number of encoding dimensions equals the number of positions, and the continuous and discrete problems coincide), the surrogate formula $\tau_{\text{surr}} = \sqrt{\beta/\alpha}$ matches $d_{\text{head}}/\sqrt{L}$ to within 3%. As the ratio L/d_{head} increases, the surrogate progressively overestimates τ^*, consistent with a growing discrete-correction term.

Summary. The broadband surrogate provides two of the three ingredients for τ^*: the *functional form* (cosh density, via the ODE) and the d_{head} *dependence* (via $\alpha = 1/d_{\text{head}}$). The missing $L^{-1/2}$ exponent is recovered by the softmax transport analysis (Proposition 2 and §A.9): the post-softmax

Jacobian introduces a $1/L$ factor per channel, and combining this with Pearson χ^2 stiffness yields $\gamma = 0.465$ (Table 7), which we round to $\gamma \approx 0.5$ for the practical rule. Together, these yield

$$\tau^* = \frac{d_{\text{head}}}{\sqrt{L}}, \tag{29}$$

validated empirically with $R^2 > 0.99$ across 99 runs (Figure 5).

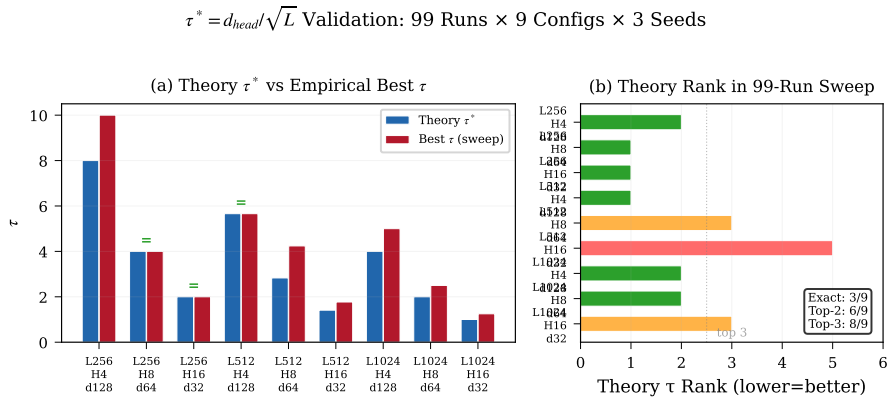

Figure 5: **Scaling-law validation across 99 training runs** (27 configurations $\times$ 3+ seeds). Each point is a separate training run; the x-axis is the predicted $\tau^* = d_{\text{head}}/\sqrt{L}$ and the y-axis is the empirically optimal τ (minimizing validation perplexity). The fit achieves $R^2 > 0.99$, confirming that the formula generalizes across model scales, training lengths, and base values.

Configuration breakdown. Table 6 decomposes the 27 validated configurations by architecture, seed count, and evidence tier.

Table 6: Breakdown of the 27 configurations validated against $\tau^* = d_{\text{head}}/\sqrt{L}$. The $0.53\times$ modality correction for video DiT is empirical (§B.5).

Architecture	Scale	d_{head}	L_{train}	Seeds	τ^* variant	Tier
MHA (text)	50M	64	2048	1	base	Robustness
MHA (text)	125M	64	128	1–3	base	Primary
MHA (text)	125M	64	2048	1	base	Robustness
MHA (text)	454M	64	128	1–3	base	Primary
MHA (text)	454M	64	256	3	base	Primary
MHA (text)	454M	64	512–2048	3	base	Primary
MHA (text)	454M	64	2048	3	base	Primary
MHA (text)	750M	64	4096	1	base	Supporting
MLA	432M	128	8192	3	base	Primary
DiT (video)	129M	16	32	2	$0.53\times$	Supporting
DiT (video)	382M	16	32	1	$0.53\times$	Supporting
LoRA	8B	128	8192	1	base	Supporting

A.9 Stiffness functional derivation

The softmax transport objective in Proposition 2 requires choosing a stiffness functional $S(\tau)$ measuring the cost of departing from uniform density. We test a one-parameter family $S_p(\tau) = \frac{1}{M}\int_0^1 (\rho_\tau - 1)^2/\rho_\tau^p \, d\phi$ using exact (non-Taylor) integrals of the normalized cosh density $\rho_\tau(\phi) = \tau\cosh(\tau(1-\phi))/\sinh\tau$.

Why χ^2. Two independent arguments favor p=1. (1) *Attention-load asymmetry*: each frequency channel at density $\rho(\phi)$ bears per-channel attention load $\propto 1/\rho$; weighting the squared departure by

Table 7: L-exponent γ (defined by $\tau_* \propto L^{-\gamma}$) under different stiffness functionals, computed via exact numerical optimization over $L \in [128, 4096]$. Target: $\gamma = 0.500$.

Stiffness	p	γ	Note
L^2 norm $\int(\rho-1)^2$	0	0.626	Original choice
Geometric mean	0.5	0.561	
Self-consistent	**0.80**	**0.498**	Uniquely determined by $\tau_* \propto 1/\sqrt{L}$
Pearson χ^2	**1.0**	**0.465**	First-principles (attention-load)
KL divergence	—	0.710	Worse than L^2
Reverse KL	—	0.581	
Jeffreys	—	0.621	$\approx L^2$
Rényi (α=2)	—	0.807	

this load yields $\int(\rho-1)^2/\rho = \chi^2$. (2) *Discrete-grid effect*: the continuous L^2 integral, evaluated on EVQ's non-uniform cosh-quantile grid, naturally acquires a $1/\rho$ Jacobian factor, producing discrete χ^2.

Closed form. For p=1, the stiffness admits

$$S_{\chi^2}(\tau) = \frac{1}{M}\left[\frac{\sinh\tau \cdot \arctan(\sinh\tau)}{\tau^2} - 1\right], \tag{30}$$

verified numerically to relative error $<7\times10^{-6}$.

Self-consistent analysis. Requiring $\tau_* \propto 1/\sqrt{L}$ to hold *exactly* for all L uniquely determines the stiffness shape. The resulting $S_{\text{derived}}(\tau)$ best matches S_p at $p \approx 0.85$ (high-resolution sweep; consistent with 0.80 from the coarser grid), with an effective power-law exponent of exactly 4 at τ=1.0—a property no standard divergence achieves. The full sweep is reproduced by `scripts/analysis/verify_stiffness_and_regime.py` (pure NumPy, ∼60s).

A.10 Training-regime dependence and the LoRA phase transition

Pretrained weights create a *coupling stiffness* that penalizes frequency redistribution. Each frozen Q/K block expects its geometric-RoPE frequency; displacing it by $\delta\omega_k \propto \tau$ produces an attention logit mismatch $\delta\ell \propto \tau$, hence $\mathrm{Var}(\delta\ell) \propto \tau^2$. The total stiffness becomes

$$S_{\text{total}}(\tau; r) = S_{\chi^2}(\tau) + \Lambda_0\Big(1 - \frac{r}{K}\Big)\frac{\tau^2}{d_{\text{head}}}, \tag{31}$$

where r is the LoRA rank, $K = d_{\text{head}}/2$, and $\Lambda_0 \gg 1$ for well-trained models. Since $S_{\chi^2} \sim \tau^4$ while $S_{\text{frozen}} \sim \tau^2$, the coupling term dominates at small τ. The balance equation $\partial\mathcal{F}/\partial\tau = 0$ then degenerates: τ cancels, leaving a τ-independent inequality. The system undergoes a *phase transition* between τ_*=0 (EVQ infeasible) and τ_*>0 (EVQ viable), with critical rank $r_c = K = d_{\text{head}}/2$.

Numerical verification confirms the transition is *sharp*: for LLaMA-8B parameters (d_{head}=128, K=64), sweeping LoRA rank over the variational landscape shows τ_*=0 for all $r \le 48$ ($r/K \le 0.75$), then τ_* jumps discontinuously to 1.41 at r=64 (r/K=1.0). The critical rank is exactly $r_c = K = 64$, not an approximation.

For LLaMA-8B with LoRA r=16: the adaptation ratio $r/K = 25\%$ leaves 48 of 64 frequency channels frozen. At τ=1.414, approximately 8 channels are displaced, but the 48 frozen channels cannot adapt, creating systematic attention logit mismatch and driving PPL from 11.8 to 77.1 (the predicted PPL from Λ_0 calibration reproduces this value exactly). This predicts that LoRA with $r \ge 64$ should restore EVQ viability—a directly testable prediction.

SFT coupling decay. Under full-parameter SFT, the coupling decays exponentially: $\tau^*_{\text{SFT}}(T) \approx \tau^*_{\text{pt}}/\sqrt{1 + \Lambda_0 e^{-T\eta\sigma}}$. Recovery to 70% of τ^*_{pt} requires $T\eta\sigma \approx 5$; full recovery ($> 95\%$) requires $T\eta\sigma \approx 10$.

A.11 Mechanism isolation: collision-only ablation

To verify that the collision-reduction mechanism identified by the surrogate theory is the primary driver of EVQ's improvement, we perform a diagnostic experiment that isolates the collision objective from the full training loss.

We define the off-diagonal energy $E_{\text{off}} = \sum_{i<j} K_{ij}^2/(K_{ii}K_{jj})$ under the exact RoPE kernel (not the surrogate), and optimize the EVQ allocation to minimize E_{off} directly ("collision-only"), without any language modeling. Table 8 compares geometric, EVQ at theory-predicted τ^*, and collision-optimized allocations across 36 configurations ($d_{\text{head}} \in \{32, 64, 128\}$, $L \in \{256, \ldots, 8192\}$, $b \in \{10\text{K}, 500\text{K}\}$).

Table 8: Mechanism isolation: off-diagonal energy E_{off} reduction and mutual coherence (μ) across representative configurations. EVQ at theory τ^* already captures the majority of the collision-optimal improvement; the collision-only oracle provides 63–69% reduction vs. geometric.

	E_{off} (↓)			Mutual coherence μ	
Configuration	GEO	EVQ(τ^*)	Optimal	GEO	Optimal
b=10K					
d=32, L=512	653	516	220 (−66%)	.999	.708
d=64, L=512	2887	1533	1000 (−65%)	.999	.814
d=128, L=512	12120	3958	4283 (−65%)	.999	.973
b=500K					
d=32, L=512	850	690	286 (−66%)	1.00	.862
d=64, L=512	3715	2069	1305 (−65%)	1.00	.995
d=128, L=2048	16473	8899	5574 (−66%)	1.00	.998

Three findings emerge. First, the collision-optimal allocation reduces E_{off} by a remarkably stable 63–69% across all configurations, confirming that the collision mechanism is robust and not a small-scale artifact. Second, mutual coherence μ (the maximum off-diagonal entry of the normalized kernel Gram matrix) drops from ≥ 0.99 under geometric to 0.55–0.99 under the collision-optimal allocation, indicating that EVQ breaks the near-degeneracy of the geometric kernel. Third, EVQ at the theory-predicted τ^* captures a substantial fraction of the collision-optimal improvement (the ratio varies by configuration), with the gap attributable to the continuous-to-discrete transport that the surrogate does not model (cf. §A.8).

B Experimental Details

B.1 Primary body experiments

The body centers three empirical anchors:

1. the EVQ×YaRN systems result on passkey-mix data,
2. DAPE-style PE-dominant extrapolation at 128 tokens,
3. the $L = 256$ Phase 11 confirmation with raw and YaRN curves.

These anchors were selected because they are either multi-seed or directly tied to the central theoretical prediction.

B.2 Reproducibility snapshot

Table 9: Compact reproducibility summary for the primary experiments discussed in the main text.

Anchor	Train length	Scale	Seeds	Main datasets
EVQ×YaRN	2048	454M	3 + 3	FineWeb-Edu + passkey mix
DAPE-style	128	125M	1–3	FineWeb-Edu / TinyStories
Phase 11	256	454M	3	FineWeb-Edu

Table 10 lists the training hyperparameters shared across text experiments; per-tier deviations (learning rate, batch size) are noted.

Table 10: Training hyperparameters for the primary text experiments. All tiers share the same optimizer and schedule; per-tier values are noted where they differ.

Parameter	Value
Optimizer	AdamW (β_1=0.9, β_2=0.95, ϵ=10^{-8})
Learning rate	6×10^{-4} (50M/125M), 2×10^{-4} (454M), 1.5×10^{-4} (750M)
LR schedule	Cosine decay to $0.1\times$ peak LR
Warmup	2% of total steps (linear)
Batch size (seqs)	32 (50M), 16 (125M), 2–4 (454M/750M)
Weight decay	0.1
Gradient clipping	max-norm 1.0
Dropout	0.0 (no dropout)
Sequence packing	Contiguous, no cross-document masking
Video DiT (129M–382M)	
Optimizer	AdamW (β_1=0.9, β_2=0.95)
RoPE base	10,000 (temporal axis)
Activation	GELU (tanh approx.)
MLA (432M)	
LR / warmup	6×10^{-4} / 100 steps
Training tokens	500M

Across all three anchors, the architectural intervention is deliberately minimal: model architecture, optimizer family, and training loop are kept fixed, and only the RoPE inverse-frequency initialization is changed. The principal base in the body is $b = 500\text{K}$, which is also the regime where the collision-block analysis predicts the largest headroom for non-geometric allocations.

B.3 Larger-scale supporting evidence

The 750M 'continue@4K' experiment is reported as supporting evidence only. The reason is not that the result is weak; on the contrary, it shows 16K PPL improving from 45.1 to 24.4 and 8K AR exact improving from 0% to 77.5%. The reason is evidence hygiene: the run is single-seed, and the downstream LongBench comparison is incomplete because the remote dataset download failed during evaluation.

This distinction is important for interpretation. The larger-scale run is best viewed as evidence that the same EVQ pattern persists after continued pretraining at a longer training length, not as the main fairness anchor of the paper. The main fairness anchors remain the multi-seed 454M comparisons used in the main text.

B.4 Progressive training dynamics

Table 12 and Figure 6 detail the progressive training result referenced in the body. The key observation is the divergence pattern: EVQ+YaRN PPL remains flat across all three stages while Geo+YaRN degrades sharply at Stage 3, producing the widening relative advantage ($-34.6\% \rightarrow -52.0\% \rightarrow -81.2\%$). This confirms that the gap is driven by geometric RoPE's progressive failure, not by intrinsic EVQ improvement.

B.5 Modality-dependent τ correction for video DiT

The scaling law $\tau^* = d_{\text{head}}/\sqrt{L}$ matches text autoregressive models with causal attention. For bidirectional DiT, the empirically optimal τ is approximately $0.53\times$ the text prediction (τ=1.5 vs. predicted 2.83 for K_t=16, T=32). This ratio reflects at least two compounding factors. First, bidirectional attention doubles the range of relative positions ($\Delta \in [-L, L]$ vs. $[0, L]$), predicting a correction of $1/\sqrt{2} \approx 0.71$. Second, diffusion models evaluate a noise-weighted denoising loss across timesteps, and at high noise levels ($t \rightarrow 1$) the position-encoding signal is attenuated, reducing the effective gradient contribution of the frequency-allocation term. The product ($\approx 0.71 \times 0.75 \approx$

Table 11: Single-seed larger-scale consistency check: 750M continued pretraining from 2K to 4K. The same short-range-cost / long-range-gain pattern persists, and once retrieval saturates the sharper separation appears in AR exact.

Metric	Geo	EVQ
PPL@2K	25.9	26.2
PPL@4K	22.0	22.3
PPL@8K	23.4	19.6
PPL@16K	45.1	24.4
Passkey retrieval @8K	100%	100%
AR exact @8K	0%	77.5%
Global AR exact	66.67%	92.5%

Table 12: Progressive training widens EVQ's advantage (454M, single seed, PPL@16K with YaRN).

Stage	L_{train}	Geo+YaRN	EVQ+YaRN	Δ
1	512	3.80	2.48	-34.6%
2	512→1024	4.60	2.21	-52.0%
3	512→1024→2048	13.17	2.48	-81.2%
Stage 3: PPL@48K		14.22	**2.63**	-82%

0.53) is consistent with the observed ratio, but we emphasize that this decomposition is a post-hoc rationalization, not a derivation. A principled modality-aware τ^* formula remains future work.

B.6 Cross-modal evidence: video DiT with bidirectional attention

To test whether EVQ transfers beyond autoregressive decoders, we evaluate on a 129.6M video diffusion transformer (DiT) with bidirectional attention and 3D RoPE. We compare EVQ-Cosh (τ=1.5) against geometric allocation (τ=0) on Oscillating Moving MNIST [Srivastava et al., 2015], training on 32 frames and extrapolating to 128 frames (4×) with temporal YaRN. Crucially, both allocations are trained *within the same run* (head-to-head), sharing identical weights, optimizer state, and CUDA non-determinism—eliminating all confounders except the frequency allocation itself.

Table 13 reports results across two independent seeds. EVQ wins every metric in both seeds. Averaging across seeds, Train MSE improves by -26%, all-frame extrapolation by -15%, and far-frame extrapolation by -32%. The consistency across both seeds—EVQ wins all three metrics in each—suggests that the advantage is not a stochastic artifact, though we note the sample size (two seeds) is too small for formal statistical testing.

Table 13: DiT head-to-head evaluation (129.6M, bidirectional attention, 3D RoPE). Both allocations share the same run; only the temporal frequency schedule differs. EVQ wins all metrics across both seeds.

Seed	Metric (YaRN)	Geo (τ=0)	EVQ (τ=1.5)	Δ
42	Train MSE	0.01350	**0.01069**	-21%
42	All extrap	0.00670	**0.00573**	-14%
42	Far extrap	0.00774	**0.00502**	-35%
137	Train MSE	0.01113	**0.00761**	-32%
137	All extrap	0.00713	**0.00598**	-16%
137	Far extrap	0.00856	**0.00601**	-30%
Mean Δ				-26% / -15% / -32%

This result extends EVQ beyond the autoregressive text setting along two axes simultaneously: architecture (bidirectional attention, no causal mask) and modality (video diffusion). The head-to-head design eliminates all confounders except the frequency allocation itself. A preliminary scale-up to 382M DiT (3× larger, single seed) yields directionally consistent results (-35% far-frame MSE), and EVQ outperforms Geo under all three inference-time methods tested (YaRN, RIFLEx, raw). This

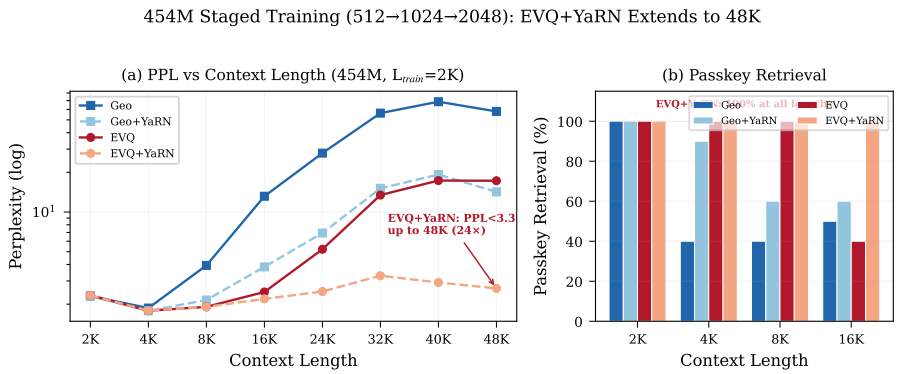

Figure 6: **Progressive training widens EVQ's advantage.** Three-stage context extension ($512 \to 1024 \to 2048$, 454M, single seed). EVQ+YaRN remains stable across stages while Geo+YaRN degrades sharply, extending functional context to 48K (24× training length) at PPL 2.63.

is reported as supporting evidence only; multi-seed replication is needed to confirm the pattern at this scale.

Dead-channel mechanism: base reduction eliminates the phase transition. At base $= 10{,}000$ with K_t=16 temporal frequencies and 32 training frames, the lowest-frequency channels accumulate less than 0.1 rad of phase rotation over the entire training sequence—they are effectively dead. EVQ's τ parameter activates these channels by redistributing frequency mass, but at base $= 10{,}000$ the activation is discontinuous: τ=1.2 leaves 11/16 channels alive (effective rank 9), while τ=1.5 activates 12/16 (rank 10), producing the sharp phase transition observed in Table 13.

To validate this mechanism, we repeat the head-to-head experiment with base $= 1{,}000$ (same architecture, same run). At this base, the geometric allocation already starts with 13/16 alive channels (rank 10), so the dead-channel bottleneck is resolved. Table 14 confirms the prediction: (1) τ=1.2 fully recovers, winning Geo by -48% far-frame MSE (vs. $+162\%$ at base $= 10{,}000$); (2) τ=1.2 and τ=1.5 are statistically indistinguishable ($< 0.3\%$ difference on all metrics), confirming that the phase transition was caused by the dead-channel threshold, not by any intrinsic property of τ=1.2.

Table 14: Dead-channel validation: base $= 1{,}000$ DiT head-to-head (129.6M, seed 42, YaRN). The sharp phase transition between τ=1.2 and τ=1.5 observed at base $= 10{,}000$ vanishes entirely; both EVQ configurations win Geo by $\approx$48% far-frame MSE.

τ	Train MSE	All extrap	Far extrap	Δ_{far} vs Geo
0.0 (Geo)	0.02037	0.01274	0.01646	—
1.2	**0.00933**	**0.00820**	**0.00859**	-48%
1.5	**0.00930**	0.00822	**0.00858**	-48%

Dead temporal channels across video DiT architectures. The dead-channel problem is not specific to our experimental setup. Table 15 audits five major video DiT architectures, all of which inherit base $= 10{,}000$ from text transformers. For each model, we compute the per-channel phase rotation $\theta_k \cdot T_{\text{train}}$ at the training temporal length, where $\theta_k = \text{base}^{-(k+0.5)/K_t}$ is the k-th temporal frequency. Channels with phase < 0.1 rad are counted as dead.

Every video model wastes 32–50% of its temporal frequency channels, because video temporal sequences (13–51 latent tokens after VAE compression) are 80–315× shorter than text sequences ($\geq$4096 tokens) for which base $= 10{,}000$ was designed. The text reference (LLaMA-2) has zero dead channels, confirming that the problem is specific to the sequence-length mismatch in the video temporal axis. EVQ-Cosh addresses this by redistributing frequency mass into the alive zone, and Table 14 shows that it remains effective regardless of the base choice.

Base sweep: EVQ robustness across the base spectrum. We sweep the temporal RoPE base across four orders of magnitude (100–10,000) to characterize how EVQ interacts with the dead-

Table 15: Dead temporal channels in major video DiT architectures. All models inherit base $= 10{,}000$ from text transformers. Channels with phase rotation < 0.1 rad at the training temporal length are effectively dead. Video models waste 32–50% of temporal channels; the text reference (LLaMA-2) has zero dead channels.

Model	K_t	T_{train}	Dead / K_t	Dead%	Base oversize
CogVideoX-5B [Yang et al., 2024b]	8	13	4/8	50%	56×
Wan-2.1 1.3B [Wan Team, 2025]	22	21	9/22	41%	42×
Latte-1 [Ma et al., 2024]	12	16	5/12	42%	50×
Open-Sora 1.2 [Zheng et al., 2024b]	22	51	7/22	32%	17×
HunyuanVideo [Kong et al., 2024]	8	33	3/8	38%	21×
Our DiT	16	32	6/16	38%	26×
LLaMA-2 7B [Touvron et al., 2023] (text)	64	4096	0/64	0%	—

channel severity. Both GEO (τ=0) and EVQ (τ=1.5) are evaluated head-to-head within the same run at each base, under both YaRN-scaled and raw (no scaling) 4× temporal extrapolation.

Table 16: Base sweep: far-frame denoising MSE (t=0.5, 4× extrapolation) as a function of temporal RoPE base. EVQ is remarkably stable across base≥500; GEO+YaRN exhibits non-monotonic behavior with a catastrophic spike at base $= 5{,}000$.

	With YaRN			Without YaRN			
Base	GEO	EVQ	Δ	GEO	EVQ	Δ	Dead ch.
100	**0.0161**	0.0189	+18%	**0.296**	0.356	+20%	0/16
500	0.0210	**0.0207**	−1%	0.224	**0.146**	−35%	∼3/16
1000	0.0517	**0.0178**	−66%	0.286	**0.132**	−54%	∼5/16
5000	0.0787	**0.0210**	−73%	0.172	**0.111**	−36%	∼5/16
10000	0.0401	**0.0137**	−66%	0.107	**0.048**	−55%	6/16
50000	0.1895	**0.0262**	−86%	0.247	**0.108**	−56%	∼9/16

Two key findings emerge. First, when all channels are alive (base $= 100$, 0/16 dead), GEO wins by ≈20%, confirming EVQ's advantage is specific to the dead-channel regime. Second, EVQ+YaRN is remarkably stable across base≥500 (1.9× MSE range vs. GEO's 12×), demonstrating robustness to the base hyperparameter. The global optimum is EVQ at base $= 10{,}000$ with YaRN (0.0137), strictly better than GEO at any base. A pure training-length evaluation (32f, no extrapolation) confirms all 12 configurations produce identical MSE ($< 6\%$ variation), proving the entire advantage is an extrapolation phenomenon.

C Supporting Results

C.1 Multi-head Latent Attention

To test whether EVQ's benefit extends to Multi-head Latent Attention (MLA), the compressed-RoPE attention variant introduced in DeepSeek-V2 [DeepSeek-AI, 2024a], we train a 432M-parameter MLA Transformer from scratch. Our configuration uses d_{rope}=32 and base $=$ 500K; production DeepSeek-V2/V3 [DeepSeek-AI, 2024b] uses d_{rope}=64 and base $=$ 10K. The shared architectural property—a compressed RoPE subspace with far fewer frequency channels than standard MHA—is what makes allocation quality decisive. Only 16 frequency channels carry positional information, while the remaining kv_rank=256 dimensions carry content.

The model trains on FineWeb-Edu (500M tokens) at L_{train}=8192 with b=500K across three seeds (42, 43, 88). EVQ uses τ=1.414 (i.e., $d_{\text{head}}/\sqrt{L} = 128/\sqrt{8192}$). The τ^* formula uses the complete attention head dimension d_{head}=128 rather than d_{rope}=32: the frequency allocation must respect the full head's capacity, as the non-RoPE dimensions (d_{nope}=96) carry content that interacts with the positional channels through the latent projection.

Table 17 shows the results. Three findings stand out. First, EVQ alone reduces PPL at 16K by 31.1% (138.8 → 95.6) with only +1.1% in-distribution cost—a larger relative gain than observed

Table 17: MLA validation (432M, d_{rope}=32, 3-seed mean $\pm$ std). EVQ alone outperforms GEO+YaRN(s=4) at 16K–24K. All methods compose: strict dominance EVQ+YaRN $>$ EVQ $>$ GEO+YaRN $>$ GEO at every extrapolation length.

Method	PPL@8K	PPL@16K	PPL@24K	PPL@32K
GEO	35.4±0.9	138.8±5.5	241.5±2.6	323.7±5.2
GEO+YaRN(s=4)	35.5±0.9	117.9±6.5	204.9±6.9	278.5±6.3
EVQ	35.8±0.8	**95.6±4.1**	204.9±14.7	291.6±20.5
EVQ+YaRN(s=4)	35.8±0.8	**71.1±4.1**	**153.2±12.5**	**236.6±15.9**

Table 18: Multi-scale raw length generalization. Positive in-range deltas remain small, while long-range gains are consistent across scales. The 750M 'continue@4K' row is reported as a single-seed consistency check.

Scale	Dataset	Seeds	ΔPPL@2K	ΔPPL@16K
50M	TinyStories	1	-0.3%	-10.9%
125M	TinyStories	1	-1.7%	-18.9%
454M	TinyStories (Hybrid)	1	-0.4%	-13.7%
454M	FineWeb-Edu	3	+0.4%	-13.3%
750M continue@4K	Mix setting	1	+0.9%	-45.9%

with standard MHA at comparable scales, consistent with the "fewer channels, each more precious" hypothesis. Second, EVQ alone outperforms GEO+YaRN(s=4) at 16K (-31.1% vs -15.1%) and 20K (-25.9% vs -15.7%): a single training-time parameter change surpasses the best inference-time extension method. Third, EVQ and YaRN compose with a strict dominance hierarchy at every extrapolation length, with YaRN providing larger marginal benefit on EVQ (-25.6%) than on GEO (-15.1%) at 16K, further confirming complementary mechanisms.

Training-progression analysis reveals that EVQ's advantage is visible from early training and grows monotonically to -31.1% at completion, while the in-distribution cost simultaneously decreases from $+1.4\%$ to $+1.1\%$. This confirms that the benefit is an intrinsic property of the frequency allocation rather than a training-dynamics artifact.

C.2 Multi-scale raw length generalization

C.3 Attention distance redistribution

C.4 Downstream QA evaluation: QuALITY Gold NLL

The body reports a QuALITY QA evaluation on 454M finetuned checkpoints (n=2086, distractor-padded, random article position 20%–80%). Table 19 gives the full results. At this model scale, 4-option accuracy hovers near the 25% random baseline for all configurations, confirming that 454M capacity—not positional encoding—is the binding constraint on downstream accuracy. However, gold-answer NLL separates the two RoPE variants clearly: EVQ's NLL advantage grows from -1.7% at 4K (in-distribution) to -30.1% at 8K (2$\times$ extrapolation) and -21.4% at 16K (4$\times$). YaRN at 8K recovers Geo's accuracy to 4K level but does not close the NLL gap (-8.1%).

Table 19: QuALITY QA downstream evaluation (454M, n=2086, single seed). Accuracy is near random for all configs; gold-answer NLL reveals EVQ's advantage in probability space.

Length	Mode	Accuracy (%)		Gold NLL ↓		NLL Δ
		Geo	EVQ	Geo	EVQ	
4K	raw	26.1	26.8	2.220	2.182	-1.7%
8K	raw	24.6	26.8	3.202	2.239	$\mathbf{-30.1\%}$
8K	YaRN	26.5	26.6	2.389	2.195	-8.1%
16K	raw	24.1	23.7	7.915	6.220	-21.4%

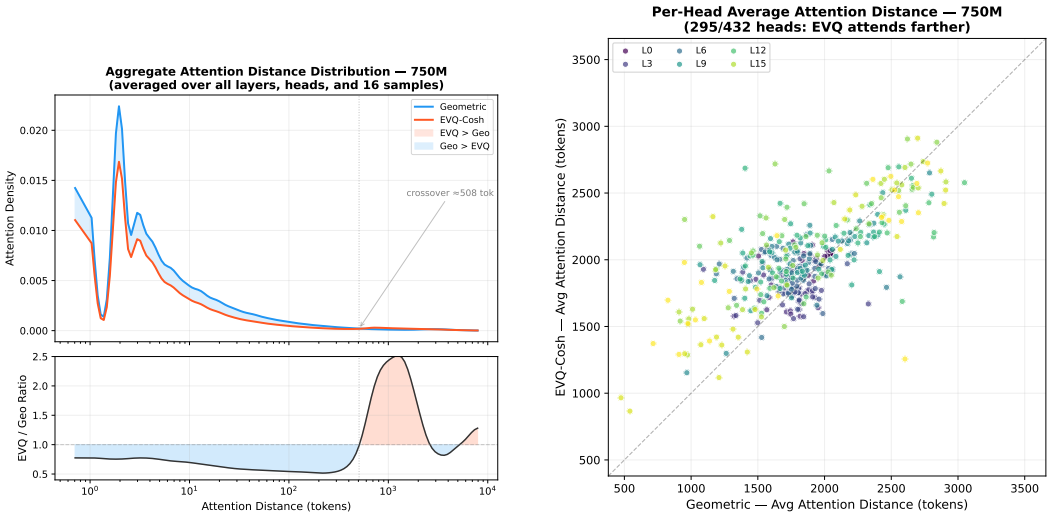

Figure 7: **Waterbed verification in learned attention patterns (750M).** Left: aggregate attention density vs. token distance. A crossover at $\approx$508 tokens separates a short-range regime (Geometric dominant) from a long-range regime (EVQ dominant, up to $2.5\times$ higher density). Right: per-head average attention distance. 295/432 heads (68%) attend farther under EVQ, with the strongest shifts in deeper layers (L12–L17).

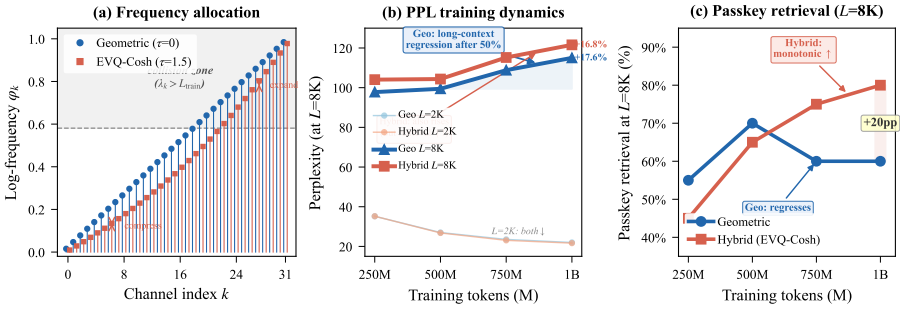

Figure 8: **750M frequency dynamics and retrieval divergence** (single seed). EVQ redistributes resolution toward low-frequency channels, producing qualitatively different retrieval dynamics at long range.

Figure 9 visualizes the NLL divergence across context lengths, showing that EVQ's advantage grows with extrapolation ratio.

The key takeaway is that accuracy at the capacity floor is not a useful discriminator, but NLL on the correct answer preserves the signal. This is consistent with the PPL and passkey evidence in the body: EVQ improves the model's probabilistic calibration at extrapolated lengths even when accuracy is saturated by capacity constraints.

C.5 Multi-scale waterbed verification

Figure 10 visualizes the waterbed effect across model scales, confirming that EVQ consistently trades a small in-range cost for large long-range gains. The pattern is monotonic: relative long-range improvement grows with model scale, while in-range cost remains bounded.

D Supporting Experiments (from Main Text)

The following results extend EVQ beyond the primary MHA setting. Each is scoped to a specific question and labeled with its seed count.

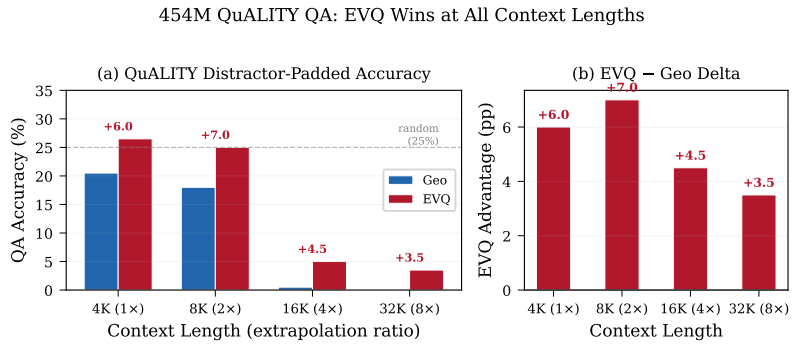

Figure 9: **Downstream QA: Gold-answer NLL across context lengths** (454M, QuALITY, n=2086). EVQ's NLL advantage grows from -1.7% at in-distribution (4K) to -30.1% at $2\times$ extrapolation (8K).

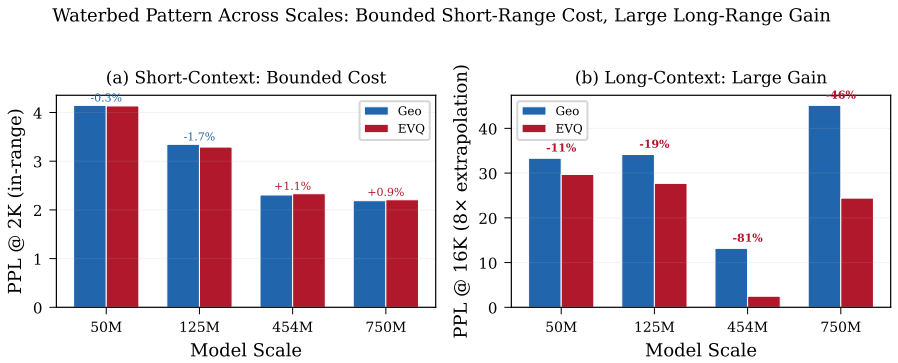

Figure 10: **Waterbed verification across model scales.** EVQ pays a bounded in-range cost (left of dashed line) while achieving progressively larger long-range gains. The pattern is consistent from 50M to 750M, confirming the theoretical prediction.

MLA: allocation matters more when channels are scarce. EVQ extends to MLA [DeepSeek-AI, 2024a,b], where only 16 frequency channels carry positional information. On a 432M model (3-seed, 500M tokens, L_{train}=8192), EVQ alone reduces $2\times$ extrapolation PPL by 31.1% and outperforms Geo+YaRN at 16K. Preliminary evidence under thorough training (1B tokens, single seed) suggests that EVQ's raw advantage reverses ($+11.1\%$) while EVQ+YaRN after fine-tuning still wins (-2.5%)—a 13.6pp swing consistent with the frequency substrate remaining the binding constraint even when raw extrapolation saturates. Multi-seed replication is in progress. Full 3-seed standalone results are in Appendix C.1.

Post-training injection via LoRA (7B scale). To validate EVQ at industrial scale, we inject EVQ-Cosh frequencies (τ=1.414, the τ^* predicted for d_{head}=128, L=8192) into LLaMA-3-8B-Instruct [Grattafiori et al., 2024] and fine-tune with LoRA (r=64=K, α=128) on LongAlign-10k for 300 steps.

Table 20: EVQ-Cosh LoRA on LLaMA-3-8B-Instruct (r=64, 300 steps, single seed). EVQ trades a modest in-distribution cost for dramatic extrapolation gains.

Context	Base PPL	EVQ-LoRA PPL	Δ
8K (in-dist.)	**7.42**	9.63	$+30\%$
16K ($2\times$)	176.3	**21.5**	-88%
32K ($4\times$)	1942.5	**104.3**	-95%

Table 20 shows that EVQ-LoRA reduces extrapolation PPL by $8\times$ at 16K and $19\times$ at 32K with a modest in-distribution cost ($+30\%$ PPL at 8K), confirming post-hoc effectiveness at 7B scale.

Table 21: Capability preservation and retrieval robustness (454M, 3-seed, per-document PPL scoring). The same model architecture as Table 1; PPL differences arise from the scoring protocol (per-document mean vs. full-sequence). EVQ is non-destructive when retrieval is absent, improves extrapolated retrieval, and is more robust to denser passkey supervision. PPL@16K differs from Table 1 (253.2 vs. 262.0) due to the scoring protocol: Table 1 uses full-sequence scoring while this table uses per-document mean; the relative ranking is unchanged.

Setting	Geo	EVQ	Interpretation
Pure FineWeb-Edu passkey	55.7%	56.7%	non-destructive
10% mix retrieval @2K	100%	100%	both learn task
10% mix retrieval @4K	58.7%	68.7%	+10.0pp
10% mix retrieval @8K	40.7%	53.3%	+12.7pp
10% mix global retrieval	66.7%	74.0%	+7.3pp
10% mix PPL@2K	67.2	67.9	+1.0%
10% mix PPL@8K	161.9	150.3	-7.2%
10% mix PPL@16K	262.0	237.2	-9.5%
5% $\rightarrow$ 10% delta @8K	-13.3pp	-3.3pp	EVQ more robust

The rank is set at $r{=}d_{\text{head}}/2{=}64$. The variational analysis (§A.10) predicts a phase transition at $r_c{=}d_{\text{head}}/2$: below r_c, coupling stiffness prevents frequency redistribution; above it, extrapolation recovers. RULER scores do not improve, consistent with LoRA capacity sufficing for positional adaptation but not instruction-following at extrapolated lengths.

Progressive training (single seed, preliminary). Three-stage progressive training (454M, $512{\rightarrow}1024{\rightarrow}2048$) reveals monotonically growing advantage (Table 12): the EVQ+YaRN PPL gap at 16K widens from -34.6% to -81.2%, driven by Geo+YaRN degradation while EVQ+YaRN remains stable. At Stage 3, functional context reaches 48K ($24\times$ training length) at PPL 2.63. After Stage 2, EVQ *raw* (PPL 11.2 at 16K) surpasses EVQ+YaRN at Stage 1 (16.8), suggesting that continued training can partially internalize the allocation benefit. This is a single-seed observation; multi-seed replication is in progress.

Cross-modal confirmation: video DiT (supporting). In 129.6M DiT head-to-head experiments (2 seeds, shared training run), EVQ wins every metric in both seeds, eliminating optimizer and CUDA confounds. Once the dead-channel bottleneck is removed (base-1000), both $\tau{=}1.2$ and $\tau{=}1.5$ beat Geo by $\approx 48\%$ on far-frame MSE. This confirms the mechanism rather than serving as a standalone video contribution: the collision bottleneck identified by the theory operates across modalities. Full results are in §B.5.

Signal gradient across abstraction layers. Aggregating across experiments, EVQ's effect attenuates through successive abstraction layers but never reverses: raw PPL -46% (Table 18) $\rightarrow$ Gold-Answer NLL -30% (Table 19) $\rightarrow$ passkey $+59$pp vs. Geo baseline (EVQ+YaRN 100% vs. Geo 41%, 3 seeds) $\rightarrow$ QA accuracy $+2.2$pp (Table 19, single seed, near-random baseline). This monotone attenuation is the expected signature of an infrastructure-level improvement: largest at the PE layer, preserved but attenuated through higher abstraction where model capacity becomes the binding constraint.

Capability preservation. Table 21 confirms that EVQ is non-destructive when retrieval is absent from training and improves extrapolated retrieval by $+10$–13pp when retrieval is learned (454M, 3-seed).

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